STA630

Glossary

- **Action Research**

That research role where the researcher and their role is involved in the research study and is interventionist. Here research is joined with action in order to plan, implement and monitor change.

- **Alternative hypothesis**

Alternative hypothesis is a statement that expresses a relationship between two variables or indicating differences among groups. It is opposite to null hypothesis. It is usually denoted by H1.

- **Applied Research**

Applied research solves specific problems and it is conducted when decision must be made for particular real-life poblem. For example, a particular product may not be selling well and manager wants to know reasons in order to take actions.

- **Basic Research**

Basic research is conducted to generate a body of knowledge which may not be immediately utilized. It is also known as Fundamental research or Pure research.

- **Bibliography**

Bibliography is the list of all works that are relevant to the topic of interested research, arranged in alphabetical order of the last name of authors.

- **Broad Problem Area**

Broad problem area refers to the entire situation where one sees a possible need of doing research and solving a problem. It is a general area of study from where research process starts.

- **Case study**

Case study is in-depth, contextual analysis of a situation or number of specific situations i.e. cases. Case study is a problem solving technique that explores problem in as rich detail as possible. Example can be case study of highly successful organization.

- **Categorical variable**

Categorical variable is a variable that has distinct value and which cannot be divided into fractions. For example, sex; male and female. Categorical variables are also known as discontinuous or classifactory variables.

- **Causal study**

The study in which researcher wants to explain the cause of one or more problems is called causal study. For example, does smoking cause cancer? The given research question is causal study.

- **Causality**

Relationship where a change in one variable produces a change in another variable. One variable affects, influences, or determines some other variable.

- **Chi-square test statistic**

Measure of the goodness of fit between the numbers observed in the sample and the numbers we should have seen in the sample, given the null hypothesis is true.

- **Close ended question**

Closed ended questions are the questions in which respondent has to choose one of the given options in a question that specifically reflects his/her answer. Close ended questions help respondents to make quick decisions to choose among several alternatives

- **Cluster sampling**

Cluster sampling is form of probability sampling which is done when groups of elements have heterogeneity among members within.

- **Coding**

Coding is assigning of numbers or symbols to the data before entering it into the computer. The assigned numbers and symbols make statistical analysis easy.

- **Communication research**

An umbrella term for market research, social research, media research and audience research combined.

- **Computer assisted personal interviewing (CAPI)**

Computer assisted personal interviewing (CAPI) is another type of Computer assisted interviews in which responses can be directly entered into the computer using microcomputers. It reduces the error and cost

- **Computer assisted telephone interviewing (CATI)**

Computer assisted telephone interviewing (CATI) is a technique to collect data in which responses gathered from telephone interviews are directly entered into the computer by using advance software. Voice recording system assists CATI programs to record interviewees' responses.

- **Concept**

An idea aimed at satisfying consumer wants and needs.

- **Confounds or confounding variables**

Extraneous causal factors (variables) that can possibly affect the dependent variable and, therefore, must be controlled.

- **Content Validity**

Establishes the representative sampling of a whole set of items that measures a concept, and reflects how well the dimensions and elements therof are delineated.

- **Data analysis**

Data analysis is the process to analyze data after collecting it. During the analysis stage, several interrelated procedures are performed to summarize and rearrange the data.

- **Data interpretation**

Data interpretation refers to making inferences pertinent to the meaning and implications of the research investigation and drawing conclusions.

- **Data presentation**

Data presentation plays an important role to interpret data and deduce results by using tables and graphs.

- **Data transformation**

Data transformation is the process of changing data from their original form to a format that is more suitable to perform a data analysis that will achieve the research objectives.

- **Deductive reasoning**

A logical process of developing specific predictions (hypotheses) from general principles. This type of reasoning moves from the general to the particular.

- **Dependent variable**

In experimental research, the dependent variable is the variable presumed within the research hypothesis to depend on (be caused by) another variable (the independent variable); it is sometimes referred to as the outcome variable.

- **Descriptive Research**

Research designed to provide a summary of some aspects of the environment when the hypotheses are tentative and speculative in nature.

- **Descriptive statistics**

Statistical methods used to describe or summarize data collected from a specific sample (e.g. mean, median, mode, range, standard deviation).

- **Desk research**

Research done by summarizing published sources - a form of secondary research.

- **Directional hypothesis (or one-tailed hypothesis)**

A hypothesis that makes a specific prediction about the nature and direction of the relationship between the independent and dependent variables.

- **Double barreled question**

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Double barreled question consists of two or more questions joined together. It makes the respondent's answer ambiguous. For example, if asked, "Are you satisfied and happy with your supervisor?"

- **Educational Research**

It is the formal, systematic application of the scientific method to the study of educational problems.

- **Empirical methods**

Research methods and data gathering techniques supported by measurable evidence, not opinion or speculation.

- **Ethics**

the rules of conduct recognized in respect of a particular class of human actions or a particular group, culture, etc.

- **Ethnographic research**

The collection of extensive narrative data on many variables over an extended period of time in a naturalistic setting in order to gain insights not possible using other types of research. For ethnographic research, observations are done at particular points of time. Data would include observations, recording and interpreting what is seen.

- **Ex post facto designs**

Ex post facto designs are designs in which there is no manipulation of independent variable in lab or field setting rather researchers only report what has happened or what is happening.

- **Experimental design**

A study in which there is random assignment of subjects to different groups so that there are no major differences between the control and comparison groups. An experimental study controls all the conditions:

•The environment

•Sample assignment (random assignment in two groups)

•Treatment/Intervention (Practice/Program)

- **Experimental group**

In experimental research the group of subjects who receive the experimental treatment or intervention under investigation.

- **Experimental research**

A research methodology used to establish cause-and-effect relationships between the independent and dependent variables by means of manipulation of variables, control and randomization. A true experiment involves the random allocation of participants to experimental and control groups, manipulation of the independent variable, and the introduction of a control group (for comparison purposes). Participants are assessed before and after the manipulation of the independent variable in order to assess its effect on the dependent variable (the outcome).

- **Extraneous variable**

A variable that interferes with the relationship between the independent and dependent variables and which therefore needs to be controlled for in some way.

- **Filter question**

A question that is asked to determine which branching question, if any, will be asked.

- **Focus Group discussion**

A generic type of qualitative research in which a small group of people provide information by discussing a topic. Some variants are the consensus group, focus group, nominal group, co-discovery conference.

- **Frequency distribution**

The number of respondents who choose each alternative answer as well as the percentage and cumulative percentage of respondents who answer.

- **Generalizability**

Generalizability refers to make research findings universally applicable. It describes the extent to which research findings can be applied to settings other than that in which they were originally tested.

- **Goodness of measure**

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Goodness of measure describes that tool developed to measure a particular concept should indeed measure it accurately and actually. The three major criteria for evaluating the tool are; reliability, validity and sensitivity.

- **Historical research**

The systematic collection and evaluation of data related to past occurrences in order to describe causes, effects, trends of those events which may help to explain present events and anticipate future events. Data is often archival-including newspaper clippings, photographs, etc.-and may include interviews.

- **Hypothesis**

A statement that predicts the relationship between variables (specifically the relationship between the independent and dependent variables). A hypothesis may be directional or non-directional.

- **Hypothesis Test**

A formal test, using probability and sampling distributions to decide which of two conflicting hypotheses should be accepted. Sometimes referred to as a significance test (where the degree of error is emphasized).

- **Independent variable**

The variable (or antecedent) that is assumed to cause or influence the dependent variable(s) or outcome. The independent variable is manipulated in experimental research to observe its effect on the dependent variable(s). It is sometimes referred to as the treatment variable.

- **Inductive reasoning**

A logical process of reasoning used to develop more general rules from specific observations; this type of reasoning moves from the specific to the more generalized.

- **Inductivism/Inductive Reasoning**

The process by which the truth of a proposition is made more probable by the accumulation of confirming evidence, a common pattern in sociological and scientific research.

- **Industrial research**

Market research in which the respondents are organizations, not consumers. Much the same as business to business research.

- **Inferential statistics**

Statistics that allow a researcher to make inferences about whether relationships observed in a sample are likely to occur in the wider population from which that sample was drawn. Inferential statistics use logic and mathematical processes in order to test hypotheses relating to a specific population based on data gathered from a sample of the population of interest.

- **Interval data**

Measurements that allow us to tell how far apart two or more objects are with respect to attributes and consequently to compare the difference between the numbers assigned. Because the interval data lack a natural or absolute origin, the absolute magnitude of the numbers cannot be compared.

- **Judgemental sampling**

Studies in which respondents are selected because it is expected that they are representative of the population of interest and/or meet the specific needs of the research study.

- **Likert scale**

Scaling technique where a large number of items that are statements of belief or intention are generated. Each item is judged according to whether it reflects a favourable or unfavourable attitude toward the object in question. Respondents are then asked to rate the attitude towards the object on each scale item in terms of a five-point category labelled scale.

- **Loaded questions**

Questions that suggest what the answer should be or indicate the researcher's position on the issue under study.

- **Mail surveys**

Data-collection method that involves sending out a fairly structured questionnaire to a sample of respondents.

- **Mall-intercept personal survey**

Survey method using a central-location test facility at a shopping mall; respondents are intercepted while they are shopping.

- **Maturation**

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Threat to internal validity; refers to changes in biology or psychology of the respondent that occur over time and can affect the dependent variable irrespective of the treatment conditions.

- **Mortality**

Threat to internal validity; refers to the differential loss (refusal to continue in the experiment) of respondents from the treatment condition groups.

- **Nominal data**

Measurement in which the numbers assigned allow us to place an object in one and only one of a set of mutually exclusive and collectively exhaustive classes with no implied ordering.

- **Non-directional hypothesis (or two-tailed hypothesis)**

A hypothesis that does not specify the nature and direction of the relationship between the independent and dependent variables.

- **Non-probability samples**

Form of sampling where there is no way of determining exactly what the chance is of selecting any particular element or sampling unit into the sample.

- **Non-response error**

Error that occurs because not all of the respondents included in the sample respond; in other words with non-response, the mean true value (on the variable of interest) of the sample respondents who do respond may be different from the entire sample's true mean value (on the variable of interest).

- **Non-sampling error**

Degree to which the mean observed value (on the variable of interest) for the respondent of a particular sample agrees with the mean true value of the particular sample of respondents (on the variable of interest).

- **Null hypothesis**

A statement that there is no relationship between the independent and dependent variables and that any relationship observed is due to chance or fluctuations in sampling.

- **Observational methods**

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Observation of behaviour, directly or indirectly, by human or mechanical methods.

- **Ordinal data**

Measurement in which the response alternatives define an ordered sequence so that the choice listed first is less (greater) that the second, the second less (greater) than the third, and so forth. The numbers assigned do not reflect the magnitude of an attribute possessed by an object.

- **Participative Research/Participant Observation**

Conscious and systematic sharing, in so far as circumstances will permit in the life activities, and on occasion, in the interests and affects of a group of persons.

- **Positivism**

This paradigm assumes that human behavior is determined by external stimuli and that it is possible to use the principles and methods traditionally employed by the natural scientist to observe and measure social phenomena.

- **Positivism/Positivist Approach**

A doctrine in the philosophy of science, positivism is characterized mainly by an insistence that science can only deal with observable entities known directly to experience. The positivist aims to construct general laws, or theories, which express relationships between phenomena. Observation and experiment will then show whether the phenomena do or do not fit the theory; explanation of phenomena consists in showing that they are instances of the general laws or regularities.

- **Primary Data**

Data collected to address a specific research objective (as opposed to secondary data).

- **Probability sampling designs**

Samples drawn in such a way that each member of the population has a known, non-zero chance of being selected.

- **Qualitative data**

Information gathered in narrative (nonnumeric) form (e.g. a transcript of an unstructured interview).

- **Qualitative Research**

Qualitative research generally examines people's words or actions in narrative or descriptive ways more closely representing the experiences of the people involved, it focuses on understandings, meanings and it takes seriously lay accounts and concepts.

- **Quantitative data**

Information gathered in numeric form.

- **Quantitative Research**

Quantitative research generally converts observations into discrete units that can be compared to other units by statistical analysis, it focuses on explanation, prediction and proof.

- **Quota sampling**

Design that involves selecting specific numbers of respondents who possess certain characteristics known, or presumed, to affect the subject of the research study.

- **Random sampling error**

Error caused when the selected sample is an imperfect representation of the overall population; therefore, the true mean value for the particular sample of respondents (on the variable of interest) differs from the true mean value for the overall population (on the variable of interest).

- **Ratio data**

Measurements that have the same properties as interval scales, but which also have a natural or absolute origin.

- **Reflexive Research**

Reflexive research involves exploring the ability of people to collectively or individually transform social structures. The concepts of 'habitus' and 'structured improvisation' capture the intersection of structure and agency, & society and the individual.

- **Reliability**

The degree to which the test consistently measures what is suppose to measure.

- **Replicability**

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A study is considered replicable if it can be repeated by another researcher. You can determine if a study is replicable by asking the following questions: •Is the description of the design detailed so that duplication of the study is possible? •Are the findings clearly described and reported? •Are the findings presented clearly and objectively? •Can the findings be enhanced with additional research?

- **Research method**

Specific procedures used to gather and analyze research data.

- **Research methodology**

Different approaches to systematic inquiry developed within a particular paradigm with associated epistemological assumptions (e.g. experimental research, grounded theory, ethno methodology).

- **Research problem**

A formal statement of question or hypothesis that implies empirical investigation.

- **Research question**

A clear statement in the form of a question of the specific issue that a researcher wishes to answer in order to address a research problem. A research problem is an issue that lends itself to systematic investigation through research.

- **Response error**

Error that occurs because respondents (who do respond) may give inaccurate answers, or a respondent's answers may be misrecorded.

- **Response rates**

The total number of respondents sent questionnaires who complete and return them, expressed as a percentage.

- **Sample**

The group of subjects from which data are collected. In other words, the group of people who participate in the study and whose data is used by the researcher.

- **Sampling frame**

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An explicit list of individuals or households that are eligible for inclusion in the sample.

- **Sampling interval**

Computed by taking n/N together with r, the first chosen element to be included in the sample, determines which elements will be included in the sample.

- **Secondary data**

Data that have been collected for another project and have already been published. Sources can be in-house or external.

- **Selection bias**

Threat to internal validity; refers to the improper assignment of respondents to treatment conditions.

- **Snowball design**

Sample formed by having each respondent, after being interviewed, identify others who belong to the target population of interest.

- **Social research**

Uses the same techniques as market research, but focuses less on business and more on public issues. See also audience research.

- **Statistical significant**

A research term used to indicate whether the results from a study are very probably not due to chance. Take, for example, a study that reports it found that student test scores improved after a reading intervention program was initiated at a school, and that these findings are "statistically significant." That means that the researcher is reasonably certain that student test scores did actually improve over time. If research findings are not statistically significant, any increase reported may be due to chance, rather than a result of the intervention.

- **Stratified sampling**

Design that involves partitioning the entire population of elements into sub-population, called strata, and then selecting elements separately from each sub-population.

- **Survey**

A method of gathering information from a number of individuals (the respondents, who collectively form a sample) in order to learn something about a larger target population from which the sample was drawn.

- **Survey Research**

This is the systematic gathering of information about individuals and collectivities, using interview, questionnaire or response methods to elicit information directly, and interpreting the resulting data by means of statistical analysis. It provides and alternative to the experimental method or participant observation and is widely used in both qualitative and quantitative research.

- **Syndicated research**

Research originated by a research company, with data sold to anybody who is interested - unlike an ad hoc survey, which is a one-off survey for an individual client.

- **Systematic methods**

Data collection methods that are consistent and scientific-i.e. that follows the scientific method, used for collecting observations.

- **Systematic sampling**

Design whereby the target sample is generated by picking an arbitrary starting point (in a list) and then picking every nth element in succession from a list.

- **Target population**

Set of people, products, firms, markets, etc., that contain the information that is of interest to the researcher.

- **Telephone surveys**

Survey that involves phoning a sample of respondents drawn from an eligible population and asking them a series of questions.

- **Theory**

In its most general sense a theory describes or explains something. Often it is the answer to 'what', 'when', 'how' or 'why' questions.

- **Validity**

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Refers to the best approximation to truth or falsity of a proposition, including propositions concerning cause-and-effect relationships.

- **Variable**

An attribute or characteristics of a person or an object that takes on different values (i.e. that varies) within the population under investigation (e.g. age, weight, pulse rate).

- **Variance**

A measure of dispersion or variability (spread), calculated by squaring the value of the standard deviation.

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